The Data Refinery — Day 5/21 Advanced SQL Architecture

Deep-Dive Architecture, Execution Internals & Performance Tuning

Author: Prasad S Hiremath **Sprint:** 21-Day Data Engineering Mastery

Part 1: Advanced SQL Terminology & Engine Internals

⚙️ 1. Hash Aggregation & Spill-to-Disk Mechanics

When executing queries with `GROUP BY` or distinct aggregations, modern distributed engines (Spark, DuckDB, Snowflake) utilize **Hash Aggregation**. Grouping keys are hashed and stored in in-memory hash buckets to update aggregate states incrementally. When unique keys exceed RAM capacity, intermediate states spill to local disk storage (Spill-to-Disk), causing severe I/O degradation.

📏 2. Window Frame Specifications: ROWS vs. RANGE

Window functions with an explicit `ORDER BY` default to a framing clause of `ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW` if omitted.

- **ROWS:** Physical row counting, ignoring duplicate values.
- **RANGE:** Logical value-based bounding. Identical values in the `ORDER BY` column are grouped together as peers, sharing identical window calculations. Essential for accurate financial moving averages.

⚡ 3. Correlated Subqueries & Vectorization Breakdown

A correlated subquery references columns from the outer query, forcing the engine to execute the subquery repeatedly for every single row evaluated in the outer dataset (row-by-row iteration). This completely destroys vectorized execution and parallel batch processing benefits, turning an $O(N)$ operation into $O(N \times M)$ or worse.

🔍 4. Predicate Pushdown & Partition Pruning

- **Partition Pruning:** In partitioned data lakes (Parquet/Delta/Iceberg stored in S3/ADLS), query engines inspect directory metadata and completely skip reading entire folder partitions that do not match filters (e.g., `WHERE dt >= '2026-01-01'`).
- **Predicate Pushdown:** Pushes filters down to physical file readers to drop irrelevant rows before shuffles.

THE DATA REFINERY

Part 2: SQL Coding Challenge & Core MCQs (1 to 3)

💻 SQL Coding Challenge: 3-Day Moving Average

SLIDING WINDOW

Scenario: Calculate a 3-day moving average (current day + 2 preceding days) for stock closing prices.

```
SELECT trade_date, stock_symbol, closing_price, AVG(closing_price) OVER ( PARTITION BY stock_symbol ORDER BY trade_date ROWS BETWEEN 2 PRECEDING AND CURRENT ROW ) AS moving_avg_3d FROM stock_prices;
```

MCQ 1: Hash Aggregation Spills

INTERNALS

What happens when unique grouping keys exceed available RAM during Hash Aggregation?

A) Intermediate hash states spill to local disk storage [CORRECT]

B) Automatic fallback to a Cartesian CROSS JOIN

MCQ 2: ROWS vs. RANGE Framing

WINDOW FRAMES

How does `RANGE` handle duplicate values in an `ORDER BY` clause compared to `ROWS`?

A) Groups logical value peers together in the frame [CORRECT]

B) Counts strict physical row offsets only

MCQ 3: Correlated Subqueries

PERFORMANCE

Why do correlated subqueries severely impact distributed data warehouse performance?

A) They execute row-by-row against outer queries, breaking vectorization [CORRECT]

B) They are restricted to string data types

THE DATA REFINERY

Part 3: Final MCQs & Bonus Q&A (6 to 10)

MCQ 4: Pushdown OPTIMIZATION

Primary benefit of Predicate Pushdown?

A) Filters rows at storage level [CORRECT]

B) Encrypts parquet files

MCQ 5: Cartesians JOINS

10k rows × 5k rows accidental CROSS JOIN?

A) 50,000,000 rows [CORRECT]

B) 15,000 rows

Bonus Expert Interview Q&A (6 to 10)

Q6: Salting Skewed Keys
Appends random hash suffixes to hot keys to balance shuffle partitions across workers.

Q7: CTE Materialization
Optimizers inline CTEs as views unless referenced multiple times or explicitly materialized.

Q8: FULL OUTER JOIN
Retains unmatched records from both left and right tables with NULL placeholders.

Q9: Default Window Frame
Defaults to ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW.

Q10: Partition Pruning Failure
Applying SQL scalar functions (e.g., `YEAR(dt)`) directly on partition columns prevents engine file skipping.

📋 **Day 5 Master Answer Key Summary**

Question Tier	Core Concepts Covered	Correct Options / Standards
MCQ 1 to 5	Hash Spills, Window Framing, Subqueries, Predicate Pushdown, Cartesians	**A, A, A, A, A**
Bonus Q6 to Q10	Salting Skew, Inlining, FULL OUTER, Default Frame, Pruning Pitfalls	**Salting, Inlining, NULL fill, Unbounded, Function wraps**

THE DATA REFINERY